Session 1 Quiz - Introduction to AI

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:56552]
Complete the following sentence: Every AI system follows a continuous cycle of Sense → Think → ______.

A. Act
B. Sleep
C. Magic
D. Human

Correct Answer: A. Act

Explanation: The Sense → Think → Act cycle is a standard way to describe how AI systems work. The AI first senses data from the world (e.g., via camera or microphone), then thinks by processing that data to make a decision or prediction, and finally acts by delivering a result such as displaying a recommendation, moving a robot arm, or speaking an answer. Sleep, magic, and human do not represent the final output step in this cycle.

Q2 [Type: MCQ | Difficulty: Easy | QID:56555]
Complete the following sentence: AI that is designed to perform ONE specific task really well, such as playing chess or recognizing faces, is called ______ AI.

A. Robot
B. Narrow
C. Broad
D. General

Correct Answer: B. Narrow

Explanation: Narrow AI (sometimes called weak AI) is designed for a single task such as chess, face recognition, or language translation. It cannot do anything outside that task. Broad and General describe the opposite (AGI, which does not exist yet). Robot refers to a physical machine that may or may not contain AI.

Q3 [Type: MCQ | Difficulty: Easy | QID:56556]
Complete the following sentence: Instead of following strict instructions, AI 'figures things out' by looking for ______ in huge amounts of data.

A. Feelings
B. Secrets
C. Instructions
D. Patterns

Correct Answer: D. Patterns

Explanation: The defining feature of modern AI is that it learns from data by discovering patterns, which are repeated similarities, correlations, or structures. For example, an AI that recognizes cats learns patterns like pointy ears and whiskers from thousands of cat photos. It does not follow hardcoded instructions for every possible case, nor does it search for secrets or feelings.

Q4 [Type: MCQ | Difficulty: Easy | QID:56558]
Complete the following sentence: To gather information from the world around it, an AI system might use ______ as its 'eyes'.

A. Microphones
B. Cameras
C. Keyboards
D. Speakers

Correct Answer: B. Cameras

Explanation: In the Sense stage of the AI cycle, cameras act as visual sensors. They capture images or video that the AI can then process. Microphones capture sound (acting as ears), keyboards are for human input, and speakers are for output (acting). Cameras are the correct choice for 'eyes'.

Q5 [Type: MCQ | Difficulty: Easy | QID: 56560]
Complete the following sentence: A program that always produces the exact same result because it follows fixed rules written by a person is a ______ program.

A. Magical
B. Intelligent
C. Traditional
D. Self-learning

Correct Answer: C. Traditional

Explanation: Traditional (or rule based) software follows explicit instructions written by a programmer. If you give it the same input, you get the same output every time. It does not learn or change. AI, by contrast, learns patterns from data and can improve over time. Magical, intelligent, and self learning are either incorrect or describe AI itself.

Q6 [Type: True/False | Difficulty: Easy | QID:56561]
True or False: AI systems have real feelings and can truly feel happy or sad based on your interactions.

A. True
B. False

Correct Answer: B. False

Explanation: AI systems do not have emotions, consciousness, or subjective experience. They process mathematical and statistical patterns and generate outputs that may sound empathetic or angry, but that is just simulation based on training data. A calculator does not feel happy when you get the answer right; similarly, no AI today actually feels happy or sad.

Q7 [Type: True/False | Difficulty: Easy | QID:56562]
True or False: Most Artificial Intelligence today lives as software code on phones or in 'the cloud' rather than inside robot bodies.

A. True
B. False

Correct Answer: A. True

Explanation: The vast majority of AI in use today is software running on smartphones, laptops, or servers in the cloud. Examples include Netflix recommendations, Google Maps traffic predictions, and spam filters. While some robots (like Roomba or Boston Dynamics) use AI, they represent a tiny fraction of AI deployments.

Q8 [Type: True/False | Difficulty: Easy | QID:56564]
True or False: If a device does the exact same thing every time without ever learning or improving, it is likely NOT an AI.

A. False
B. True

Correct Answer: B. True

Explanation: The ability to learn from data and improve over time (or adapt to new inputs) is central to what makes something AI. A simple thermostat or digital clock follows fixed rules and never changes its behavior; those are not AI. A spam filter, by contrast, learns from new emails and gets better, so it is AI.

Q9 [Type: True/False | Difficulty: Easy | QID:56566]
True or False: Alan Turing was a scientist who helped start the field of AI by asking if machines could 'think'.

A. False
B. True

Correct Answer: B. True

Explanation: Alan Turing's 1950 paper "Computing Machinery and Intelligence" proposed the Turing Test, a way to evaluate whether a machine could exhibit intelligent behavior indistinguishable from a human. That question and his work laid foundational ideas for the field of artificial intelligence.

Q10 [Type: True/False | Difficulty: Easy | QID:56567]
True or False: AI is always 100% correct, so you should never double-check the answers it gives you.

A. False
B. True

Correct Answer: A. False

Explanation: AI systems make mistakes, sometimes confidently wrong ones. They do not understand truth or meaning; they predict patterns. For critical decisions (medical diagnosis, legal advice, schoolwork), outputs should be reviewed by a human. The statement says AI is always 100% correct, which is false.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:56569]
Why is AI compared to a calculator in the lesson?

A. Because they both have physical buttons
B. Because calculators are the most advanced form of AI
C. Because they both produce results without actually understanding them
D. Because both were invented in the year 1950

Correct Answer: C. Because they both produce results without actually understanding them

Explanation: Both a calculator and an AI system manipulate symbols or numbers according to rules (explicit or learned), but neither understands the meaning. A calculator processes "2 + 2" and shows "4" without knowing what counting is. Similarly, an AI can write a poem about love without feeling love. The comparison highlights the lack of genuine understanding.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:56570]
Which of these is an example of an AI system in the 'Think' stage of its cycle?

A. A sensor feeling the temperature in a room
B. A screen showing you a 'Top Pick' movie
C. A microphone recording your voice
D. A self-driving car deciding whether to brake or turn

Correct Answer: D. A self-driving car deciding whether to brake or turn

Explanation: The Think stage involves processing sensed data and making a decision or prediction. A self-driving car analyzing camera images and deciding "brake" or "turn" is classic thinking. Sensing (A and C) is the Sense stage. Showing a top pick (B) is the Act stage (output). Deciding is the core of thinking.

Q13 [Type: MCQ | Difficulty: Medium | QID:56571]
How does an AI 'create' a brand new picture or story?

A. It remixes patterns from millions of examples it has seen
B. It has its own unique imagination and dreams
C. It waits for a human to draw it first
D. It searches the internet and steals one existing file

Correct Answer: A. It remixes patterns from millions of examples it has seen

Explanation: Generative AI (like image or text generators) does not have imagination. It learns statistical patterns from a huge training dataset (millions of images or documents) and then combines those patterns in new ways. The result looks original but is essentially a remix of what it has seen. It does not steal a single file, nor does it have dreams.

Q14 [Type: MCQ | Difficulty: Medium | QID:56572]
Which of these is a 'superpower' that humans have but AI does not?

A. Spotting patterns in data that humans might miss
B. Working 24/7 without ever needing to sleep
C. True creativity and original ideas
D. Processing millions of files every second

Correct Answer: C. True creativity and original ideas

Explanation: AI can generate novel combinations, but its outputs are based on training data patterns, not on consciousness, emotion, or lived experience. Humans can produce truly original ideas, art, and stories from internal feelings and intentionality. The other options (pattern spotting, non-stop work, fast processing) are AI strengths, not human unique superpowers.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:56573]
What happened in 1997 that caused people to 'freak out' about Artificial Intelligence?

A. The first robot was built to cook dinner
B. A computer called Deep Blue beat the world chess champion
C. The internet was invented and shared with the world
D. AI learned how to have human feelings

Correct Answer: B. A computer called Deep Blue beat the world chess champion

Explanation: In 1997, IBM's Deep Blue defeated Garry Kasparov, the reigning world chess champion. This was a major public milestone because chess was seen as a pinnacle of human intellect. People were surprised (and some alarmed) that a machine could outthink a human in such a complex strategic game. The internet existed before 1997, and AI has not learned human feelings.

Q16 [Type: MCQ | Difficulty: Medium | QID:56574]
Why is AI described as a 'Tool' rather than a 'Replacement'?

A. Because humans are always the ones in charge and responsible
B. Because it can do everything a person can do better
C. Because it is too expensive for most people to use
D. Because it is made of metal and plastic

Correct Answer: A. Because humans are always the ones in charge and responsible

Explanation: AI is a tool because it assists humans but does not take ultimate responsibility or moral agency. Humans decide how to use AI, what goals to set, and whether to trust its outputs. AI does not replace human judgment, accountability, or values. The other options are false: AI cannot do everything better, it is becoming cheap, and being metal or plastic is irrelevant.

Q17 [Type: MCQ | Difficulty: Easy | QID:56575]
Which of these apps uses AI to 'Sense' your face so it can apply a filter?

A. A physical microwave oven
B. A simple Calculator app
C. The phone's clock app
D. Snapchat or Instagram

Correct Answer: D. Snapchat or Instagram

Explanation: Snapchat, Instagram, and similar apps use facial recognition AI to sense facial landmarks (eyes, nose, mouth) in real time from the camera feed. This allows filters to track your face and overlay effects. Microwaves, calculators, and clock apps do not sense faces or use AI for this purpose.

Q18 [Type: MCQ | Difficulty: Easy-Medium | QID:56577]
What is the 'secret' to how AI learns what a cat looks like?

A. It goes to a pet store and meets real cats
B. It is programmed with a list of rules about every animal
C. It sees thousands of pictures and finds what they have in common
D. It uses magic to see inside the internet

Correct Answer: C. It sees thousands of pictures and finds what they have in common

Explanation: AI learns by being trained on many labeled examples (thousands or millions of cat photos). It finds statistical patterns, common features like pointy ears, whiskers, and fur textures, without being given explicit rules for every animal. It does not visit pet stores, follow hardcoded rules for all animals, or use magic.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:56578]
What does Google Maps use AI for?

A. To predict traffic and find the fastest route
B. To drive your car for you automatically
C. To take photos of your house every day
D. To tell you the time in different countries

Correct Answer: A. To predict traffic and find the fastest route

Explanation: Google Maps uses AI (specifically machine learning) to analyze real time location data from millions of phones, predict traffic congestion, and estimate travel times. It then suggests the fastest route. It does not drive your car (that is self driving cars, a different system), nor does it take daily house photos or primarily tell time across countries.

Q20 [Type: MCQ | Difficulty: Easy | QID:56579]
In the AI cycle, which stage is represented when Siri speaks the weather forecast out loud?

A. Act
B. Think
C. Wait
D. Sense

Correct Answer: A. Act

Explanation: The Act stage is when the AI produces an output or performs an action based on its thinking. Siri speaking the weather forecast using speech synthesis is a clear example of acting. Sensing would be listening to your question. Thinking would be processing that question and retrieving the forecast. Wait is not part of the standard cycle.